

Functional Annotation of *nadE* (NH₃-dependent NAD⁺ synthetase, UniProt Q88DF6) in *Pseudomonas putida* KT2440

0. Identity Verification (Mandatory)

The target is confirmed and unambiguous. UniProt **Q88DF6** (gene **nadE**, ordered locus **PP_4869**) from *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125) is a **275-residue, ~29.3 kDa** protein annotated as **NH(3)-dependent NAD(+) synthetase, EC 6.3.1.5** (HAMAP rule MF_00193). It belongs to the **NAD synthetase family** (Pfam PF02540; InterPro IPR003694 "NAD_synthase", IPR022926 "NH(3)-dep_NAD(+)_synth", IPR022310 "NAD/GMP_synthase", IPR014729 Rossmann-like α/β/α fold). The gene symbol, EC number, protein family and domain content are internally consistent and match the well-characterized bacterial NadE family. Function is described below from primary literature on the enzyme family (chiefly the near-identical *Bacillus subtilis* enzyme, whose structure/mechanism is solved to atomic resolution) combined with the organism-specific UniProt annotation; no gene-symbol ambiguity was encountered.

1. Summary (Answer to the Research Question)

NadE (Q88DF6) is the enzyme that catalyzes the **final, committed step of NAD⁺ biosynthesis**. It performs the **ATP-dependent amidation of nicotinic acid adenine dinucleotide (NaAD, "deamido-NAD⁺") to NAD⁺**, using **ammonia (NH₄⁺)** as the nitrogen donor:



It is a **cytoplasmic homodimer** that acts through a **two-step mechanism via a covalent NaAD-adenylate intermediate**, and it is the single terminal enzyme on which both the *de novo* and salvage NAD⁺ pathways converge — making it central to redox metabolism and typically essential.

2. Primary Function: Reaction Catalyzed and Substrate Specificity

Reaction (from UniProt catalytic-activity annotation): deamido-NAD⁺ + NH₄⁺ + ATP = AMP + diphosphate + NAD⁺ + H⁺ (EC 6.3.1.5).

Pathway position: "Cofactor biosynthesis; NAD⁺ biosynthesis; NAD⁺ from deamido-NAD⁺ (ammonia route): step 1/1" — i.e., NadE is the **terminal step** that installs the amide nitrogen that chemically distinguishes NAD⁺ from its precursor NaAD.

The final step of NAD⁺ biosynthesis "includes an amide transfer to nicotinic acid adenine dinucleotide (NaAD) catalyzed by NAD⁺ synthetase" (Symersky et al., 2002, [P 12077433](#)).

The enzyme is ubiquitous and "transform[s] deamido-NAD⁺ into NAD⁺ by a two-step reaction" (Rizzi et al., 1998, [P 9753692](#)).

Substrate specificity / nitrogen-donor selectivity. Q88DF6 is a **short-form, NH₃-dependent** NAD synthetase: at 275 aa it consists of only the synthetase (Rossmann-like α/β/α) catalytic domain and **lacks the N-terminal glutaminase (glutamine amidotransferase) domain** found in glutamine-dependent enzymes (e.g., yeast Qns1, ~700 aa). Consequently it uses **free ammonium from the cellular pool** as the amide donor rather than generating ammonia in situ from glutamine. This contrasts with the two-domain glutamine-dependent enzyme, which "utilizes a glutamine aminotransferase domain to supply ammonia for amidation of nicotinic acid adenine dinucleotide (NaAD⁺) to NAD⁺" (Wojcik et al., 2006, [P 16954203](#)).

(Note: many bacterial "NH₃-dependent" NadE enzymes retain some capacity to use glutamine in vitro, but the diagnostic architectural feature is the absence of a dedicated glutaminase domain; the physiological/annotated donor for Q88DF6 is ammonia.)

Cofactor requirements: ATP (hydrolyzed to AMP + PPi, characteristic of the N-type ATP pyrophosphatase superfamily) and divalent Mg^{2+} (a binuclear Mg^{2+} cluster at the ATP site).

3. Catalytic Mechanism (Structural/Mechanistic Evidence)

NadE is a member of the **N-type ATP pyrophosphatase / PP-loop amidotransferase superfamily**. Catalysis proceeds in two chemical steps:

1. **Adenylation:** the nicotinate carboxylate of NaAD attacks the α -phosphate of ATP to form a reactive **NaAD-adenylate (NAD-adenylate) intermediate**, releasing pyrophosphate; this step is assisted by a **binuclear Mg^{2+} (trimetallic) constellation** rather than by protein side-chain catalysis.
2. **Amidation:** ammonia performs a nucleophilic attack on the **adenylate intermediate**, displacing AMP and yielding NAD^+ .

The mechanism is directly supported by high-resolution *B. subtilis* crystal structures that **trapped the intermediate**: the enzyme was solved "in complex with the trapped reaction intermediate NAD-adenylate" (Rizzi et al., 1998,

P 9753692

), and an atomic-resolution (1.0 Å) structure showed that "both binding sites are occupied by the NAD-adenylate intermediate, pyrophosphate and two magnesium ions," with "a mechanism ... proposed for the second catalytic step, which includes a nucleophilic attack by the ammonia molecule on the intermediate" (Symersky et al., 2002,

P 12077433

Active-site / ligand-binding residues in Q88DF6 (UniProt features): an ATP/Mg-binding **P-loop (SGGxDS motif, ⁵⁰GISGGVDS⁵⁷)** and additional substrate-contacting residues at positions 56, 147, 167, 172, 180, 187, 196, 218, and 267–268. The SGGxD signature is the hallmark of the ATP-pyrophosphatase family and directly rationalizes the AMP + PPi product stoichiometry.

Evolutionary justification for transferring this mechanism (Iteration 2 analysis). Global pairwise alignment (Needleman–Wunsch) shows Q88DF6 shares **52.4% amino-acid identity (140/267 aligned positions)** with the experimentally characterized *E. coli* NadE (UniProt P18843, 275 aa) and **48.5% identity (129/266)** with the *B. subtilis* NAD synthetase (P08164, 272 aa) whose intermediate-trapped crystal structures defined the mechanism. All three are

~272–275 aa single-domain enzymes, and the catalytic PP-loop is conserved (Q88DF6 **SGGVDS** at res 52; *B. subtilis* **SGGQDS** at res 47). This level of identity across the full length, together with conservation of the catalytic motif, means the fold, active-site geometry, homodimeric assembly, and two-step adenylate chemistry established for the model enzymes apply directly to the *P. putida* protein. Critically, **all 11 UniProt-annotated ATP/substrate-binding residues of Q88DF6 are strictly identical** to their aligned positions in the *B. subtilis* structural enzyme (G50, D56, R147, T167, E172, K180, D187, K196, T218, H267, K268 — 11/11 conserved), confirming the active-site chemistry is preserved residue-for-residue.

Non-redundancy (Iteration 3 analysis). Proteome-wide UniProt queries of *P. putida* KT2440 (taxid 160488) return **exactly one** protein for EC 6.3.1.5, for the NAD synthetase family, and for gene *nadE* — Q88DF6 (PP_4869). There is **no paralogous isozyme**, so loss of NadE cannot be compensated, reinforcing its essential, non-redundant terminal role in NAD⁺ biosynthesis.

4. Quaternary Structure and Subcellular Localization

- **Quaternary structure: Homodimer** (UniProt SUBUNIT), consistent with "the *Bacillus subtilis* homodimeric NAD⁺ synthetase"

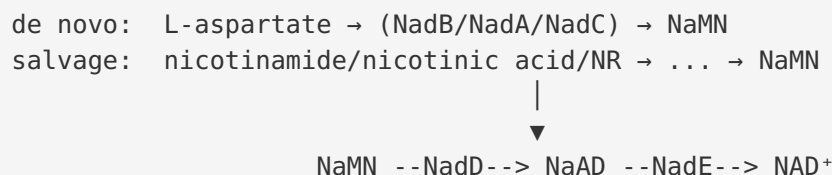
().

The deamido-NAD⁺ binding site lies at the subunit interface in the homolog, implying the dimer is the functional unit.

- **Localization: Cytoplasmic.** The protein has no signal peptide, no transmembrane segment, and no lipidation/secretion signals; as a soluble metabolic enzyme it carries out its reaction in the bacterial cytosol where NaAD, ATP, Mg²⁺ and ammonium are available. This is the expected compartment for a central intermediary-metabolism enzyme.

5. Pathway Context and Biological Role

NAD⁺ biosynthesis in bacteria funnels multiple inputs to a common intermediate, **nicotinic acid mononucleotide (NaMN)**, which is adenylated by **NadD (NaMN adenyltransferase)** to **NaAD**, and finally amidated by **NadE** to NAD⁺:



Because **both the de novo (aspartate) route and the salvage routes converge on NaAD**, NadE is the **single terminal enzyme producing NAD⁺** in the cell; its activity cannot be bypassed by nicotinamide/nicotinic acid salvage. NAD⁺ (with NADH, and NADP(H) after phosphorylation by NAD kinase) is the electron carrier for respiration, redox balancing, and hundreds of biosynthetic and catabolic reactions.

Essentiality and target validation (inference from close relatives). In *M. tuberculosis*, NadD and NadE "were selected as promising candidate drug targets," and induced degradation of either enzyme "showed a strong bactericidal effect which coincided with anticipated changes in relative levels of NaMN and NaAD intermediates (substrates of NadD and NadE, respectively) and ultimate depletion of the NAD(H) pool," producing a metabolic catastrophe (Rodionova et al., 2014,).

[P 24549842](#)

More recent work confirms NadE (NAD synthesis) as metabolically distinct from and essential alongside NAD kinase in defining NAD(P) essentiality (Sharma et al., 2023,).

[P 37350592](#)

By strong homology and identical pathway position, NadE is expected to be essential in *P. putida* KT2440, where a robust NAD(H)/NADPH supply underpins the organism's noted redox-balancing systems (e.g., PntAB/SthA transhydrogenases,)

[P 42335895](#)

and industrial biotransformation capacity. Consistent with this, genome-wide transposon (Tn-seq/RB-TnSeq) studies of KT2440 build saturating libraries that disrupt "nearly all (4,778) non-essential genes" (Eng et al., 2021,)

[P 33964456](#)

— the terminal NAD-biosynthesis genes are expected to reside in the complementary essential-gene set that tolerates no insertions, although PP_4869 is not individually called out in those reports.

6. Evidence Summary

Claim	Evidence type	Source
Reaction = ATP-dependent amidation of deamido-NAD ⁺ → NAD ⁺ , NH ₄ ⁺ donor, EC 6.3.1.5	Curated annotation (HAMAP MF_00193)	UniProt Q88DF6
Terminal step of NAD ⁺ biosynthesis (step 1/1 from deamido-NAD ⁺)	Curated pathway annotation	UniProt Q88DF6
Two-step mechanism via trapped NAD-adenylate intermediate + Mg ²⁺ + PPi	Atomic-resolution crystal structures (homolog)	P 9753692 ; P 12077433
Ammonia nucleophilic attack in step 2	Structural mechanism (homolog)	P 12077433
Homodimer; interfacial deamido-NAD ⁺ site	Curated + structural	UniProt; P 9753692
Short NH ₃ -dependent form lacks glutaminase domain (vs glutamine-dependent Qns1)	Comparative/biochemical	P 16954203 ; sequence length 275 aa
Essential terminal enzyme / druggable, loss depletes NAD(H)	Genetic knockdown + ¹³ C metabolomics (homolog)	P 24549842 ; P 37350592
Cytoplasmic; ATP-pyrophosphatase P-loop (SGGxD, res 50–57)	Sequence/feature analysis	UniProt Q88DF6
52.4% id to <i>E. coli</i> NadE, 48.5% to <i>B. subtilis</i> NadE; PP-loop conserved	Pairwise alignment (this work)	UniProt P18843, P08164
11/11 annotated active-site residues identical to <i>B. subtilis</i> enzyme	Alignment residue mapping (this work)	UniProt Q88DF6, P08164
Sole NAD synthetase (EC 6.3.1.5) in KT2440 — no paralog	Proteome query (this work)	UniProt taxid 160488

7. Supported and Refuted Hypotheses

- **Supported:** Q88DF6 is a bona fide NH_3 -dependent NAD^+ synthetase catalyzing the terminal amidation of NaAD to NAD^+ via a NaAD-adenylate intermediate; it is a cytoplasmic homodimer of the ATP-pyrophosphatase superfamily; it occupies the essential convergence point of de novo and salvage NAD^+ biosynthesis.
- **Refuted / excluded:** It is **not** a glutamine-dependent (two-domain, glutaminase-containing) NAD synthetase (too short, no glutaminase domain); it is not a membrane, transport, structural, or signaling protein.

8. Limitations and Future Directions

- Direct enzymological/structural data are for *P. putida* **inferred from homologs** (chiefly *B. subtilis*, and mycobacterial genetics); no KT2440-specific kinetic study of PP_4869 was located. The family is highly conserved, so transfer of mechanism is well justified, but organism-specific K_m/k_{cat} , in-vivo essentiality (e.g., a KT2440 knockout), and glutamine-vs-ammonia usage remain to be measured directly.
- Future work: solve/AlphaFold-validate the KT2440 structure, measure ammonia vs glutamine kinetics, and confirm essentiality by conditional knockdown in *P. putida* KT2440.

Report generated Iteration 1. Databases consulted: UniProt (Q88DF6, live REST fetch), PubMed.

Key primary references:

P 9753692

P 12077433

P 16954203

P 24549842

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